LEARNING WHAT TO CHANGE

From Visual Reasoning to Scientific Hypothesis Revision

Chimdi Walter Ndubuisi · University of Missouri REVIEW DRAFT 19 October: for collaborator review, not for print

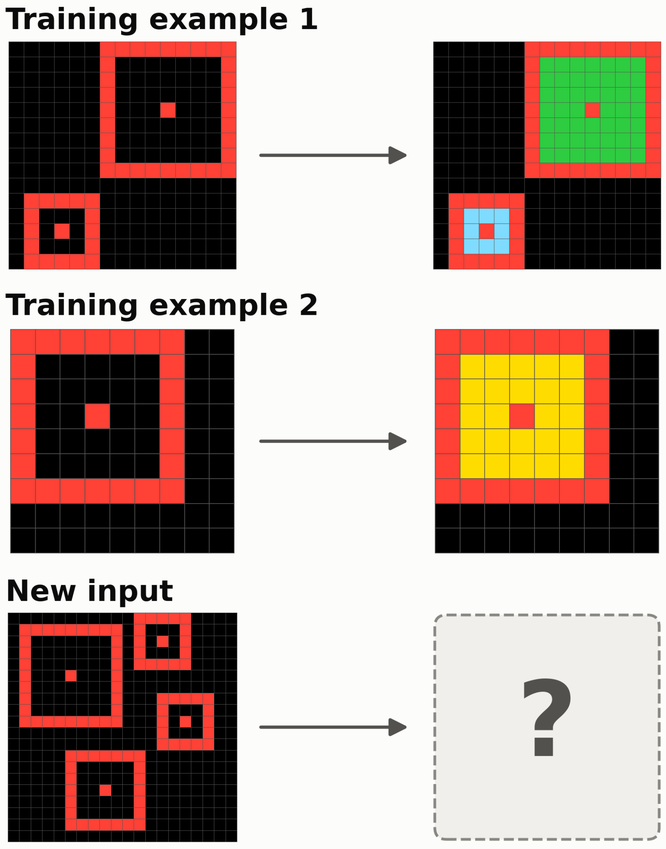
Can a reasoner recognize when its explanation is inadequate, and determine what needs to change?

Find what is missing, choose a revision, test it. A question shaped in discussions with Toni Kazic.

1 · Why reasoning fails

One rule must fit both examples and the new input.

REAL ARC-AGI-2 TRAINING TASK 00dbd492 (public)



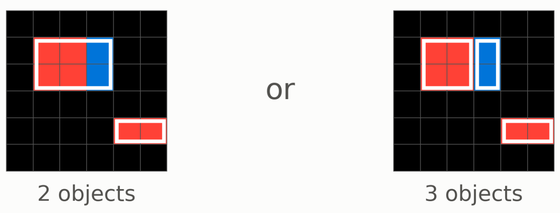
Rule: each frame's fill colour depends on its size.

Three ways an explanation fails

ILLUSTRATIVE — drawn to explain, not a CORA result

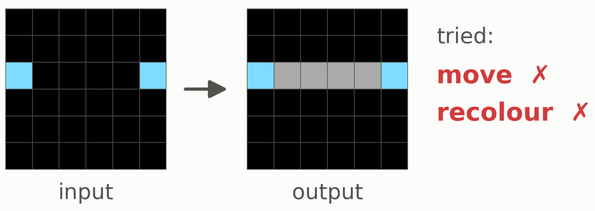
Wrong objects

The grid is split into the wrong pieces.



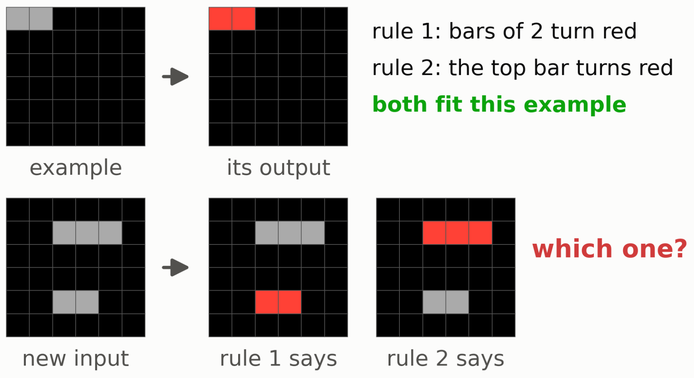
Wrong transformation

Right pieces, but no operation makes the change.



Not enough evidence

Two rules fit the example but give different answers.



Telling these apart is the research problem.

2 · What CORA has demonstrated

CORA writes rules as programs.

Fitting is not enough [DONE]

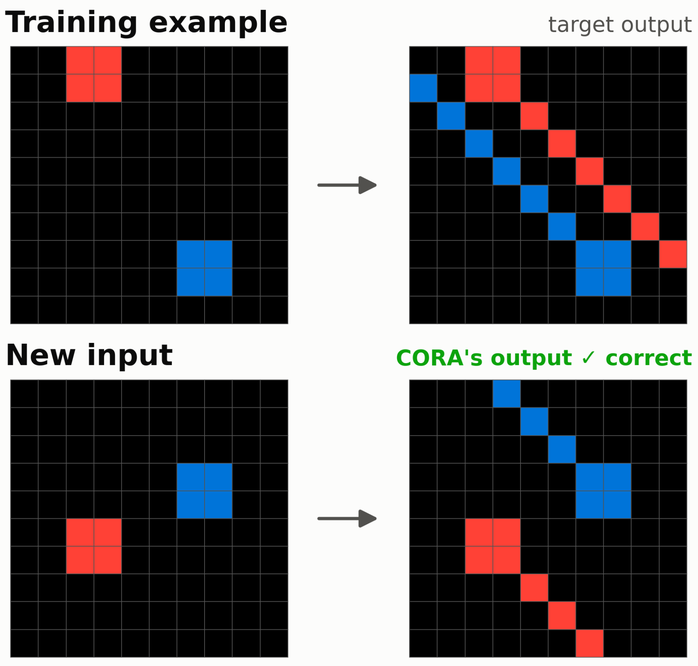
6 accepted, all correct

26 tasks: every fitted rule failed

60 exposed training tasks.

A rule CORA built [DONE, NARROW]

REAL ARC-AGI-2 TRAINING TASK ka95aa62f67d9 · recorded run



Rule: **blue** objects send a line up-left; **red** ones, down-right.

- ✓ constructed by CORA
- ✓ correct on this task
- ✓ rebuilt 3 of 3 times, one example hidden
- = also found without the failure record

An exposed training task, not a Kaggle success.

Did it generalize? [NEGATIVE]

0 / 58

new verified solutions

REAL · 58 separately selected, exposed training tasks

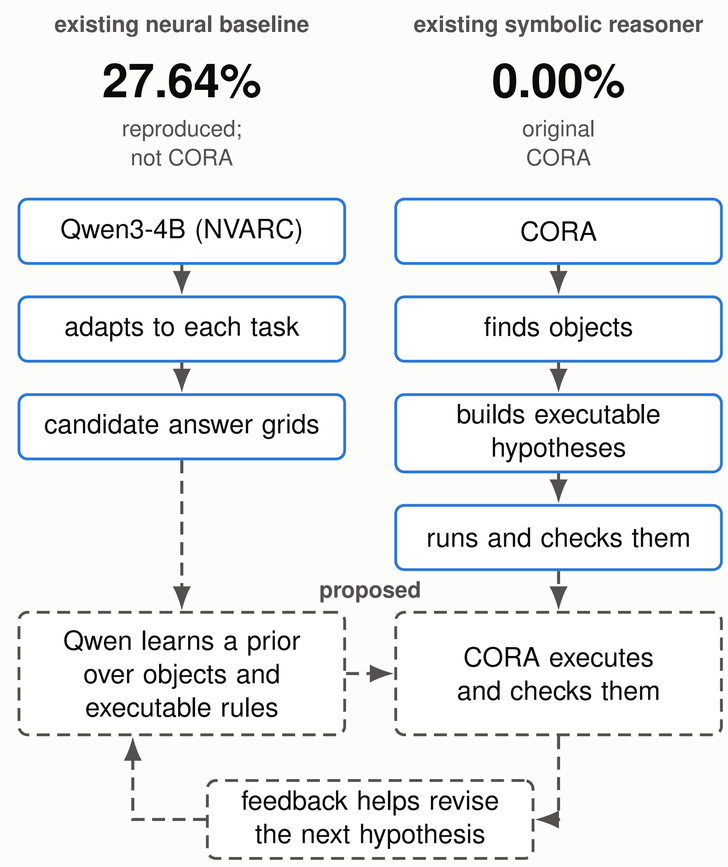
55 no rule proposed

3 rule proposed, but rejected by CORA's checks

0 new verified solutions

Every variant found 0; no wrong answers.

3 · How neural learning may help



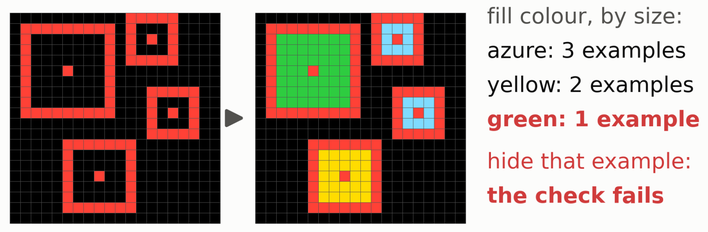
Can neural learning help a reasoner discover useful representations and rules—and can executable verification help it determine what to change when those explanations are inadequate?

Solid: existing. Dashed: proposed; the neural proposal interface has not yet been implemented or experimentally validated. Scores: official public.

CORA-v2 [DEVELOPMENT]

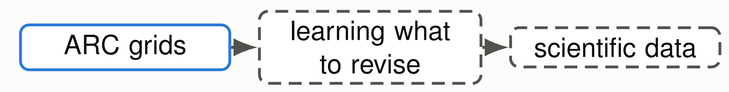
Keeps all fitting rules; flags thin evidence.

REAL TASK 00dbd492 · CORA-v2 development run
CORRECT DEVELOPMENT PREDICTION — NOT CERTIFIED



118 exposed tasks: 4 right, 1 wrong guess; none certified.

From ARC to science [FUTURE, NOT TESTED]



Proteins, omics, plant phenotypes: only the revision policy could transfer.

Building a rule is not learning which revision improves an unseen prediction. That is what we test.

existing, tested · proposed, not yet validated · [DONE] [NEGATIVE] [DEVELOPMENT] [FUTURE, NOT TESTED]

Prior work: Popper, DreamCoder, LILO, Model Discovery Agent; failing and revising is not new. Neural baseline: NVARC's published method, reproduced. Numbers: `records/CORA_SUMMIT_CLAIM_LEDGER_20261017.md` · AI coding assistance used · authorship pending review.